

Class-9 UART Class Code

```
#####

/*Master Side Code */

/*
 * File:  MasterMain.c
 * Author: sagar
 *
 * Created on 19 May, 2025, 12:03 PM
 */

#include <xc.h>
#include <stdint.h>

#define _XTAL_FREQ 20000000

void uartInit(void);
uint8_t value;

void main(void)
{
    uartInit();

    while(1)
    {
        value = PORTB;
        switch(value)
        {
            case 0xE0:
                TXREG = 'A';
                __delay_ms(100);
                break;
            case 0xD0:
                TXREG = 'B';
                __delay_ms(100);
                break;
            case 0xB0:
                TXREG = 'C';
                __delay_ms(100);
                break;
            case 0x70:
                TXREG = 'D';
                __delay_ms(100);
                break;
        }
    }
}

void uartInit(void)
```

```
{
    TRISB = 0xF0;
    PORTB = 0x00;
    OPTION_REG &= ~(0x1U << 7); // Enable Pull Up

    // UART Configuration
    TXSTA = 0x20;
    RCSTA = 0x90;
    SPBRG = 0x09;

    TRISC6 = 0; // TX pin as output
}
```

#####

/* Slave Side Code */

```
/*
 * File: SlaveMain.c
 * Author: sagar
 *
 * Created on 19 May, 2025, 12:05 PM
 */
```

```
#include <xc.h>
#include <stdint.h>
```

```
#define _XTAL_FREQ 20000000
```

```
void uartInit(void);
uint8_t RXvalue;
```

```
void main(void)
{
    uartInit();
    TRISD = 0x00; // PORTD as output
    PORTD = 0x00;

    while(1)
    {
        if(PIR1 & (0x1U << 5)) // Check RCIF flag
        {
            RXvalue = RCREG;
            switch(RXvalue)
            {
                case 'A':
                    PORTD = 0x40; // RD6 On, rest Off
                    break;
                case 'B':
                    PORTD = 0x02; // RD1 On, rest Off
                    break;
                case 'C':
                    PORTD = 0x00; // All Off
                    break;
                case 'D':
                    PORTD = 0x42; // RD6 and RD1 On
                    break;
            }
        }
    }
}
```

```
void uartInit(void)
{
```

```
TRISB = 0xF0;
PORTB = 0x00;
OPTION_REG &= ~(0x01 << 7);

// UART Configuration
TXSTA = 0x20;
RCSTA = 0x90;
SPBRG = 0x09;

TRISC7 = 1; // RX as input
}
```